A Thermodynamically-Consistent MeshGraphNet Surrogate for Transient Welding Heat Transfer

Enthalpy-State GENERIC Structure and Push-Forward Training

[Author(s)]

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Goal: a surrogate that is *thermodynamically consistent by construction*.

Governing physics and the FEM ground truth

Thickness-integrated transient heat conduction with a moving Goldak double-ellipsoid source:

$$\rho\, c_p^{\mathrm{app}}(T)\,\frac{\partial T}{\partial t} = \nabla\cdot\big(k(T)\,\nabla T\big) + Q(\boldsymbol{x}, t),$$

P_1 triangles, θ-method (backward Euler) + Picard for the nonlinearity; mixed Robin (convection + radiation) / Dirichlet boundaries.

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Dataset. 360 simulations (398 597 graph pairs); every axis covered (size, power, speed, BCs) *except trajectory topology*, which is held out $\Rightarrow$ generalization is cleanly attributable to the path.

The backbone: an Encoder–Processor–Decoder MeshGraphNet

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Accurate — but unconstrained, it can violate physics over long rollouts. We make the increment structure-preserving.

Key insight: GENERIC on *temperature* is inconsistent

GENERIC writes $\dot{\boldsymbol{z}} = \boldsymbol{L}\,\partial_{\boldsymbol{z}} E + \boldsymbol{M}\,\partial_{\boldsymbol{z}} S$, with $\boldsymbol{L}^{\mathsf{T}} = -\boldsymbol{L}$, $\boldsymbol{M}$ SPSD, degeneracies $\boldsymbol{L}\,\partial_{\boldsymbol{z}} S = \boldsymbol{M}\,\partial_{\boldsymbol{z}} E = \mathbf{0} \Rightarrow\ \dot{E} = 0,\ \dot{S} \geq 0$. Heat conduction is purely dissipative ($\boldsymbol{L} = \mathbf{0}$).

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Tempting but wrong: take $\boldsymbol{z} = \boldsymbol{T}$ and set $\Delta T_i = (\boldsymbol{L}_w \boldsymbol{\mu})_i$, $\mu_i = 1/T_i$. Then $\sum_i \Delta T_i = 0$ — but

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Fix: impose the structure on *energy* — the classical enthalpy method.

The enthalpy-state GENERIC head

Conserved state = *extensive* nodal energy $\mathcal{E}_i = h_i V_i \Rightarrow \partial E / \partial \mathcal{E}_i = 1$, so the degeneracy $\boldsymbol{L}_w \mathbf{1} = \mathbf{0}$ holds *exactly on any mesh*.

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Open system: arc source $+$ boundary losses enter as a separate non-conservative term $\boldsymbol{f}^{\mathrm{ext}}$, the only part GENERIC leaves unconstrained.

Two enablers: enriched source & conditioning

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Per-node learned modulation of the Goldak heating / Newton cooling, *zero-initialized* (starts identical to a scalar gain), softplus$(\cdot) \geq 0$, gated by q_i (cannot invent far-field energy):

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Conditioning. Raw $h \sim 10^9\ \mathrm{J/m^3}$ stalls Adam. Work in *temperature-equivalent* enthalpy $H = h/(\rho c_p)$ (units of K): a constant rescale $\Rightarrow$ conservation unchanged, gradients $O(1)$, training unstuck from epoch 1.

Aligning the objective: push-forward ($K=2$) training

Empirical lesson (repeated across the study): pushing the *single-step* objective harder (lower train MSE / short-rollout val) did *not* improve — sometimes worsened — the 1526-step inference rollout.

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K=1 recovers the classic objective; K=2 makes the operator contractive over two-step composition — enough, with the structure, to align with the long rollout while staying cheap.

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Plus standard MeshGraphNet training noise ($\sigma = 5\,\mathrm{K}$). Headline run: A100-80GB, $\sim$15 min/epoch, 80 epochs.

Out-of-distribution probe: the Archimedean spiral

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- Trajectory **never seen** in training (only straight/diagonal/sinusoid/arc).
- Plate size and speed *interior* to training ranges $\Rightarrow$ isolates *trajectory* generalization.
- 4225 nodes, 1526 steps (76.25 s) — far longer than any training window.
- Exercises genuine *re-heating*: successive arms re-heat one another (multi-pass, absent from training).

Why it is hard

Longest horizon $\times$ unseen topology $\times$ multi-pass physics — the regime where unconstrained surrogates fail.

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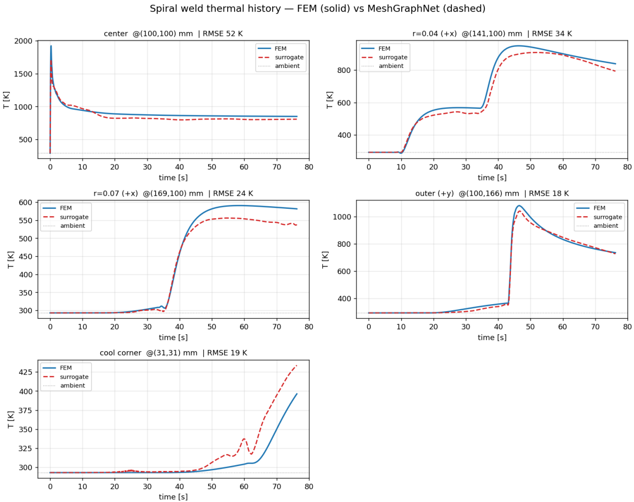

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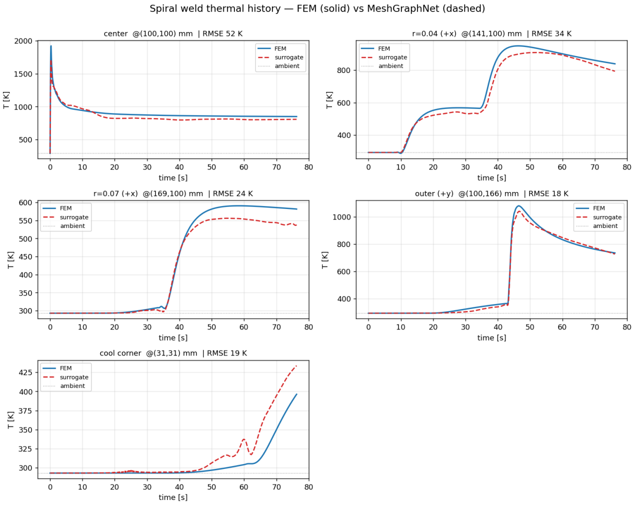

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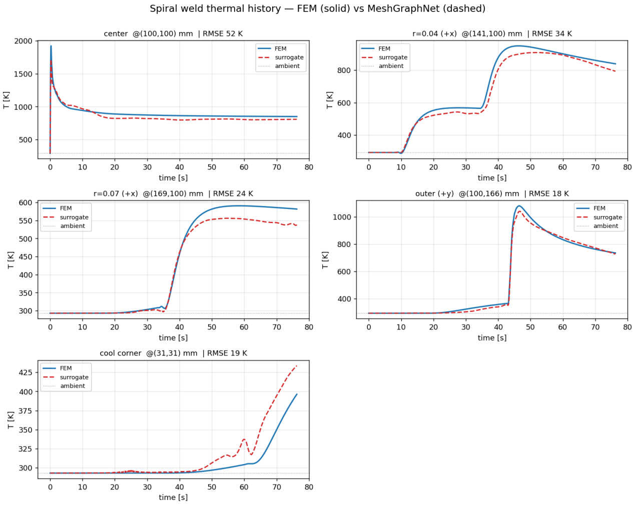

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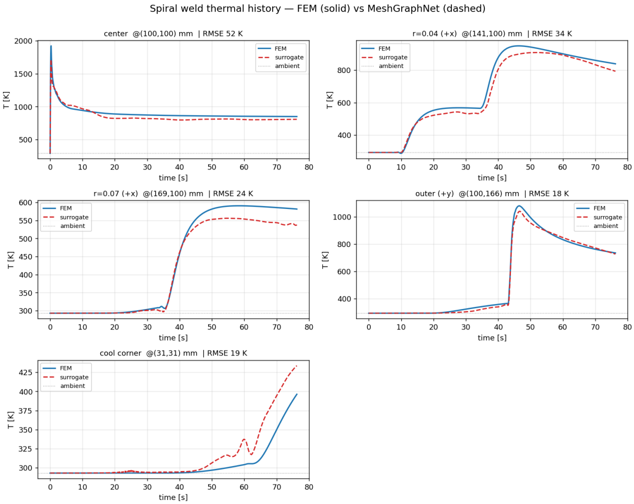

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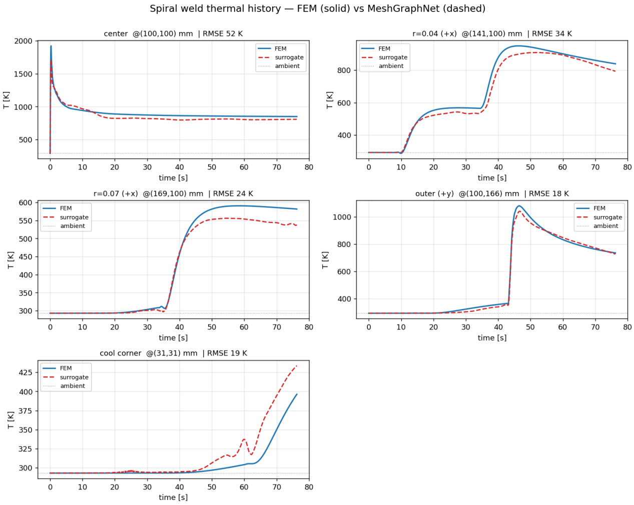

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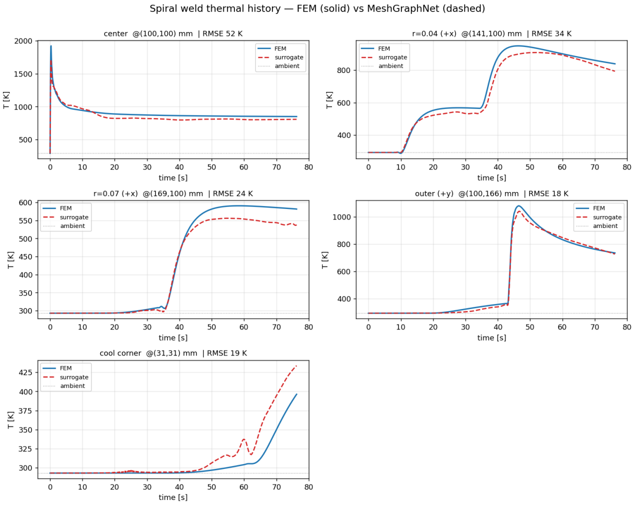

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Remaining flaw: the sharpest central peak is

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ON (enthalpy GENERIC + source)	1 375 238	34.5	**33.2**	33–61 [3.4]	287 (-6)
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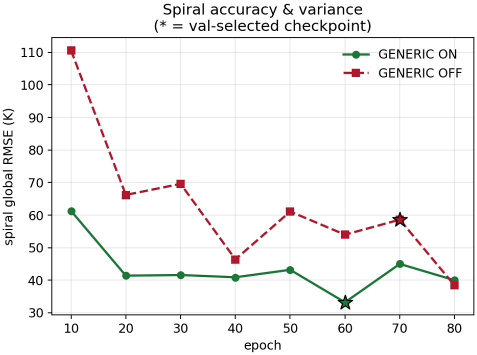
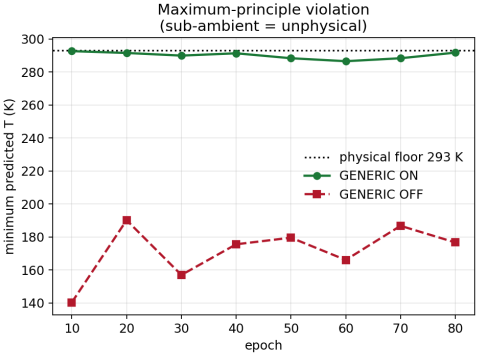
- **Selectability:** for ON the best-val checkpoint *is* the best spiral; for OFF it is not.

Ablation, visually: accurate-but-fragile vs. guaranteed





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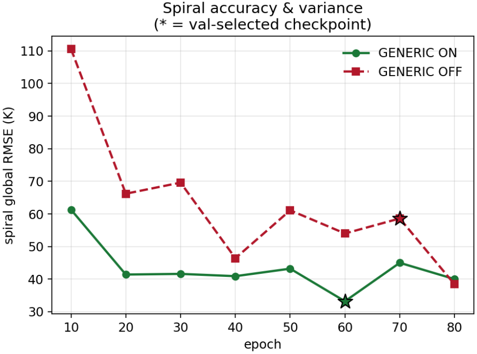

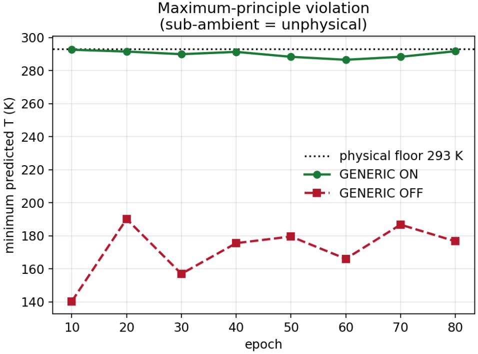

Left: spiral RMSE vs epoch — ON (tight band, ⋆ best-val coincides with best spiral); OFF swings ∼3×.
Right: min temperature — ON at the 293 K floor; OFF sub-ambient at every checkpoint.

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